KIRO

Agentic AI development environment from prototype to production

{Speaker Name}

Agenda

{ 1 }

AI is changing how software is built and operated

{ 2 }

Spec Driven Development

{ 3 }

Kiro Clients

{ 4 }

Enterprise Pricing

cloudar

AI is changing software

2024 ASSISTANTS

Helping developers write code faster

2025 AGENTS

Completing development tasks end-to-end with human in the loop

2026 AUTONOMY

Completing development tasks end-to-end with bounded independence

cloudar

The promise of agentic development

Autonomy

Agents can take on and complete increasingly challenging tasks autonomously

True Collaboration

Developers and agents work together to get more done

Higher quality

Agents handle the heavy lifting of increasing test coverage, documenting code, optimizing performance, fixing security vulnerabilities

Challenges with AI development

Scaling AI development

AI coding tools excel at small tasks but can fail with complex projects

Limited control

Existing tools make it difficult to collaborate with and manage agents

Code quality

Getting a project from proof-of-concept to production while maintaining quality control becomes increasingly difficult

What would a development experience look like if it could take full advantage of working with AI agents?

Solution: Spec-driven development

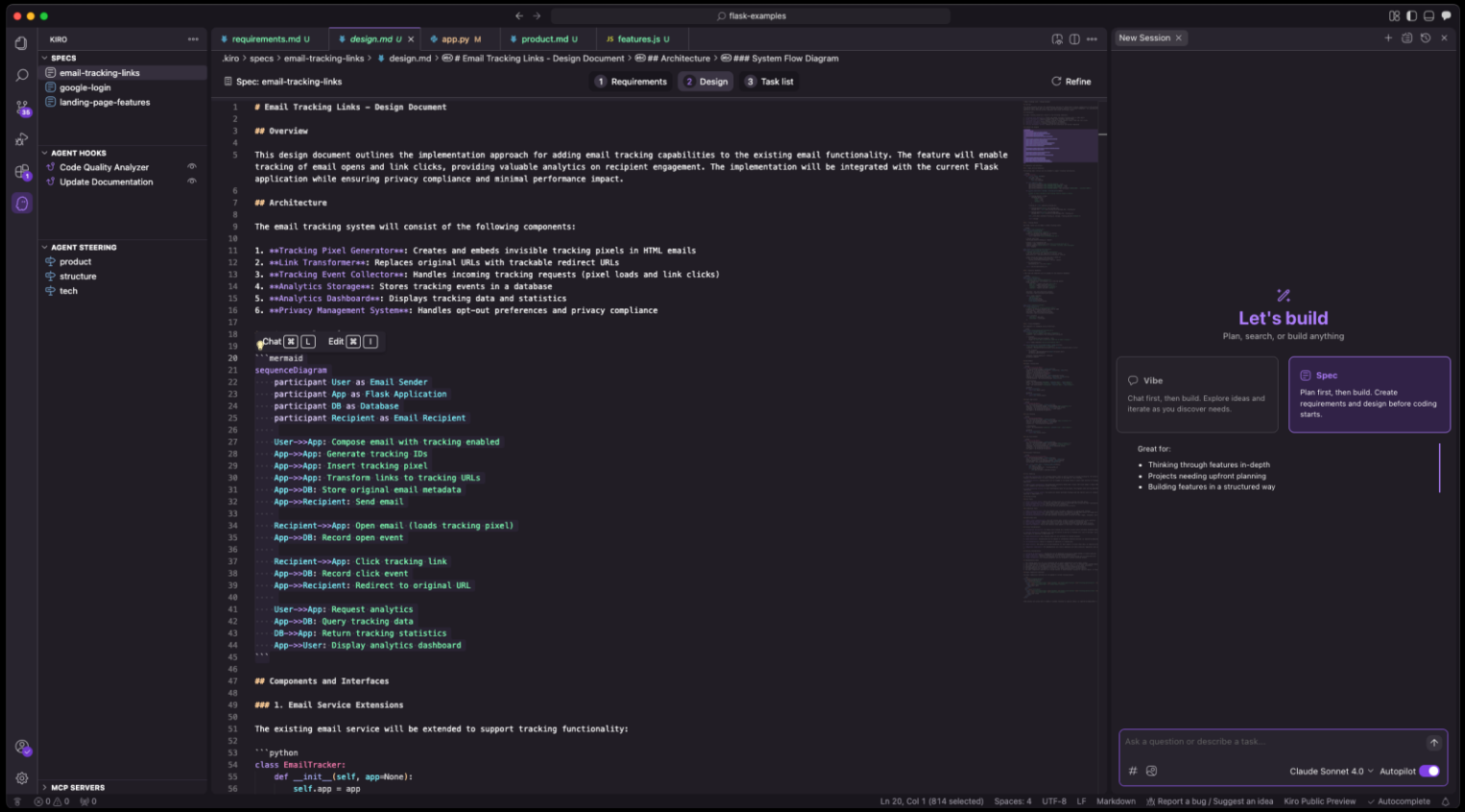
Vibe coding ships the prototype, traditional SDLC practices ship the product

Kiro helps developers and engineering teams
ship high quality software with AI agents

cloudar

KIRO IDE

The AI IDE for prototype to production



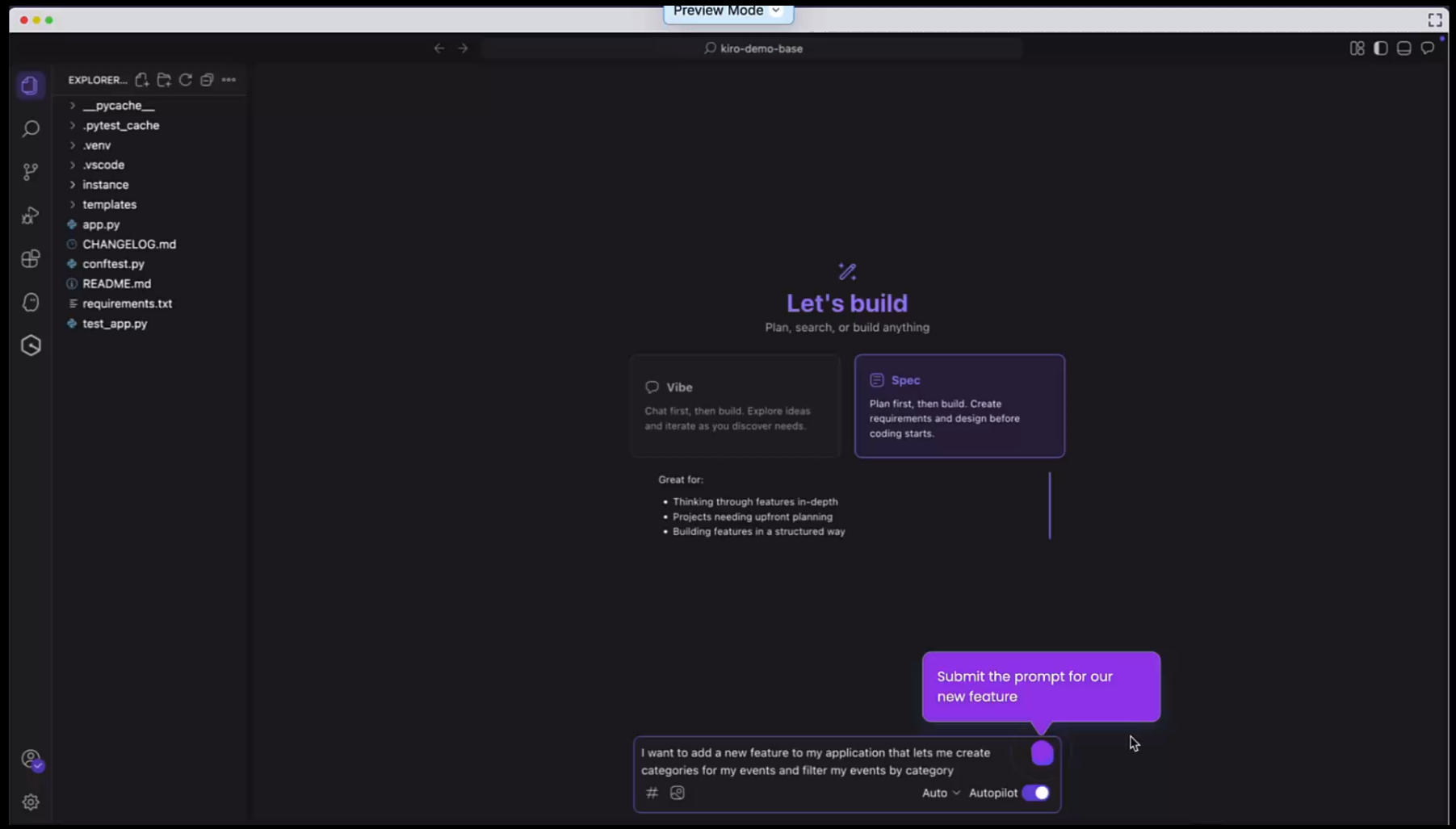
Kiro helps you do your best work by bringing structure to AI coding with spec-driven development.

cloudar

KIRO IDE

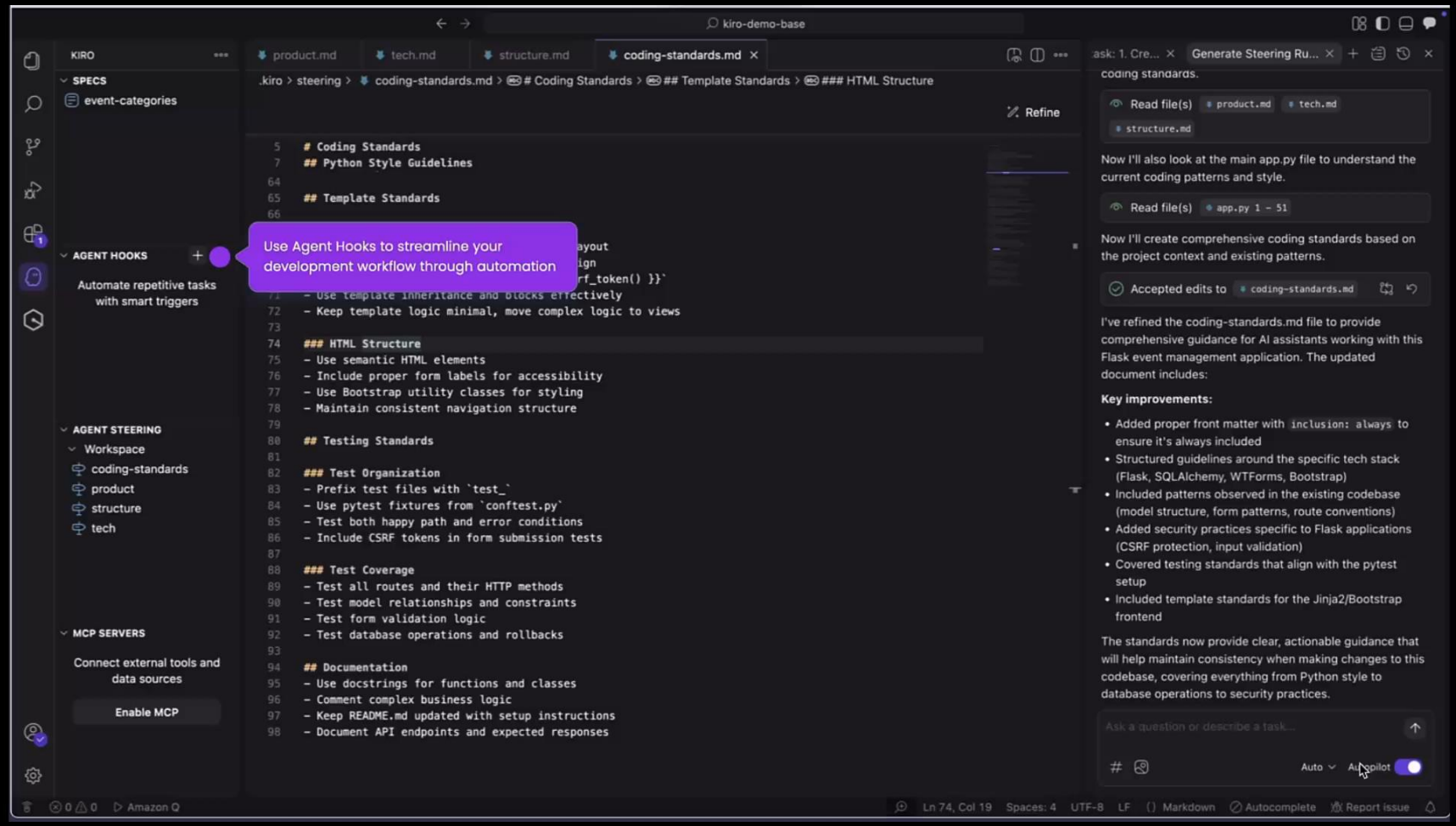
Spec-driven development

- Kiro turns your prompt into clear requirements, system design, and discrete tasks
- Iterate with Kiro on your spec and architecture
- Kiro agents implement the spec while keeping you in control



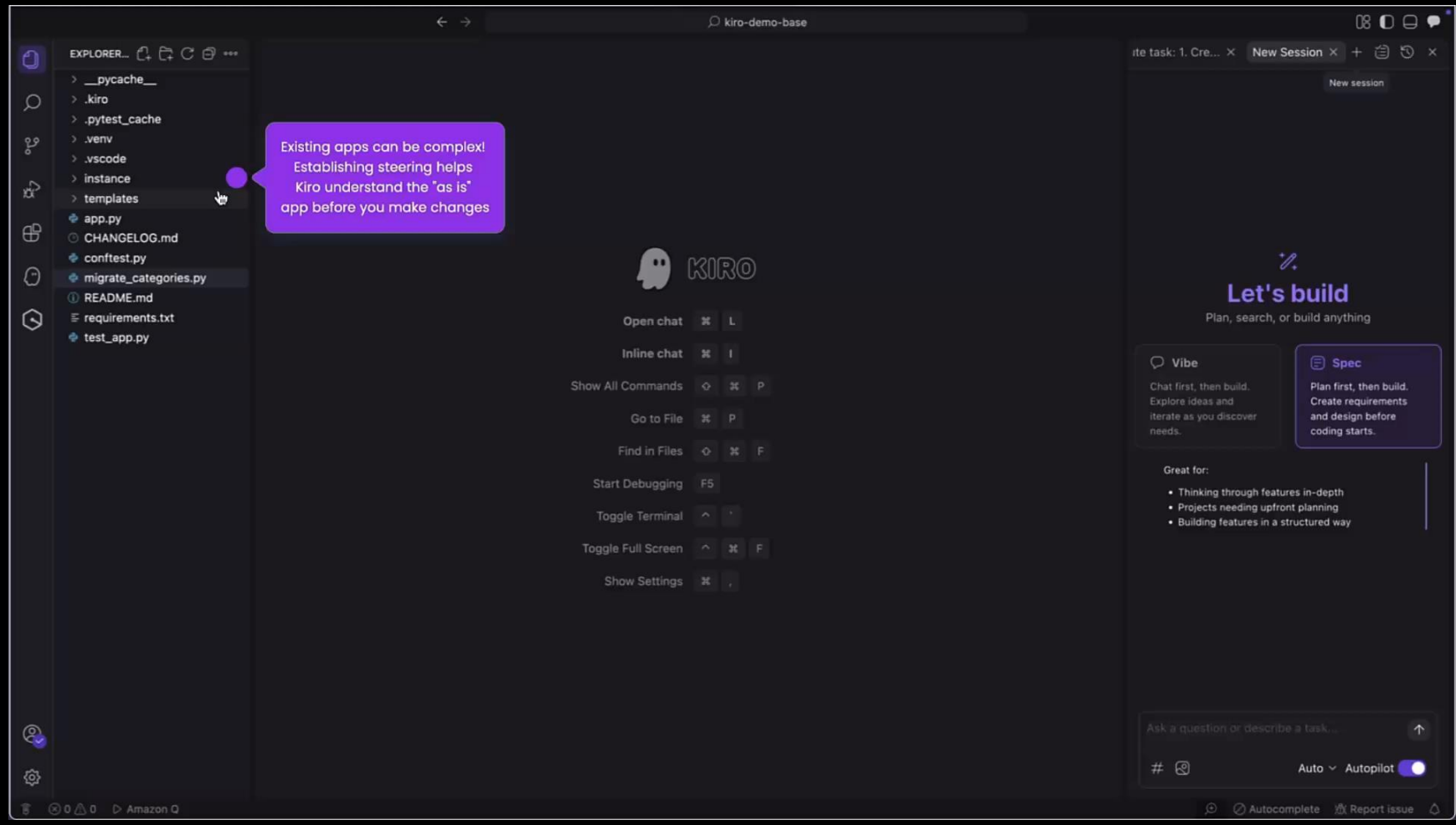
Agent hooks

- Delegate tasks to AI agents that trigger on events such as 'file save'
- Agents autonomously execute in the background based on your pre-defined prompts
- Agent hooks help you scale your work by generating documentation, unit tests, or optimizing code performance



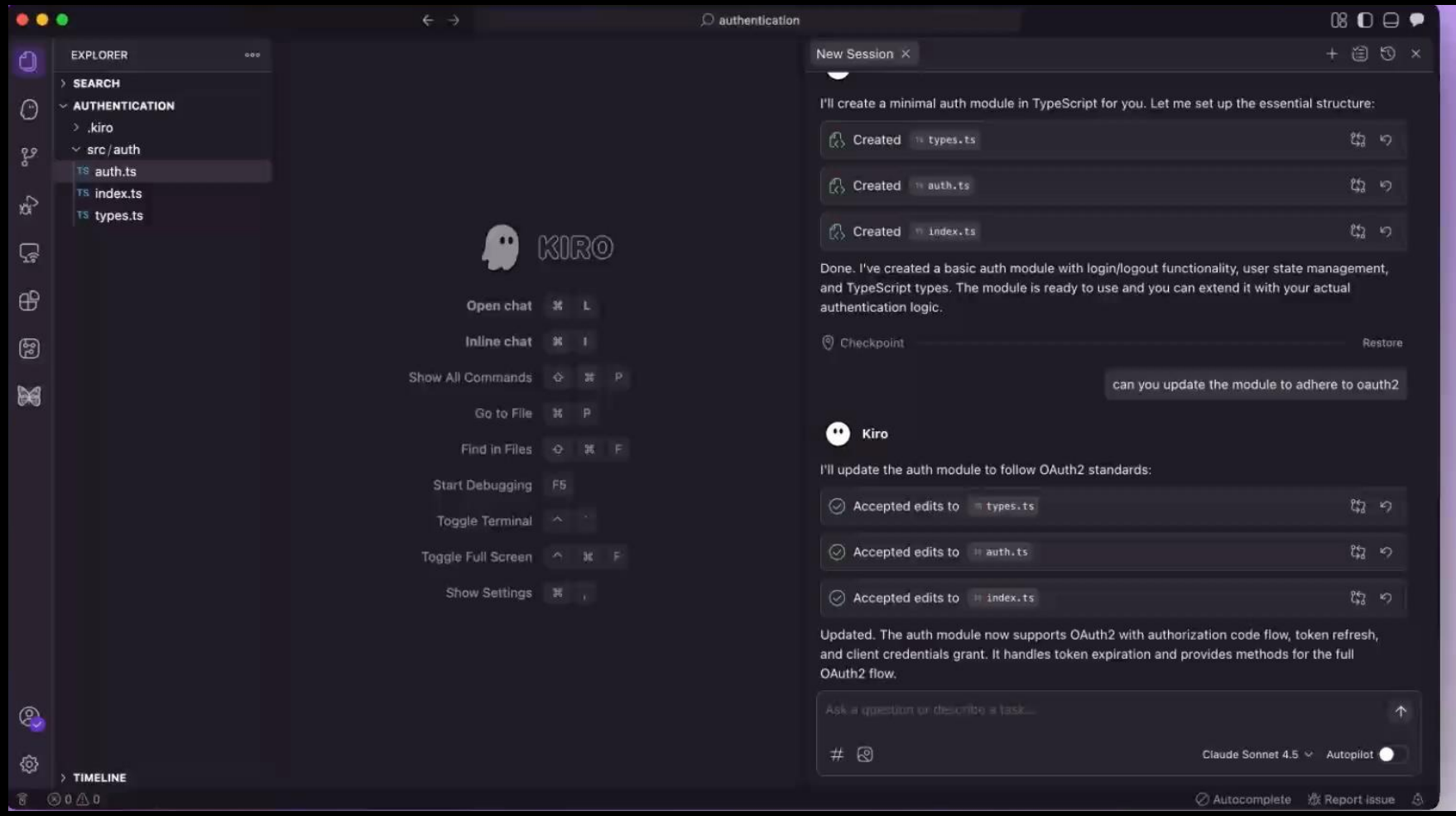
Advanced context management

- Connect to docs, databases, APIs, and more with native MCP integration
- Configure how you want Kiro agents to interact with each project via steering files
- Drop an image of your UI design, or a photo of your architecture whiteboarding session, and Kiro can use it to guide its implementation



Timeline Checkpointing

- Rollback to existing points in the execution log
- Safety net to safely explore multiple approaches to a problem
- Snapshots the contents of each modified file and then restores that snapshot when reverting to the checkpoint



Spec correctness – Property-based testing

How do you know that user intent matches the generated output?

Spec correctness

w. Property-based testing

A property looks like:

for any (x, y, …)
such that precondition(x, y, …) holds
predicate(x, y, …) is true

KIRO CLI

Kiro is now available in the terminal

```
kanbo@ac07752a88c7 kiro-demo-base % clear
kanbo@ac07752a88c7 kiro-demo-base %
```

Turn ideas into implementation with speed and precision, while staying in complete control every step of the way

cloudar

KIRO CLI

CLI Custom Agents

- Optimize Kiro for specific tasks like documentation, code reviews, and unit tests.
- Provide pre-approved tool permissions, context files and prompts
- Kiro builds upon your feedback, delivering intelligent outputs while reducing development interruptions

```
- awslabs.cdk-mcp-server

Did you know?

You can specify wait time (in ms) for mcp server loading with
kiro-cli settings mcp.initTimeout {timeout in int}. Servers
that take longer than the specified time will continue to load in the
background. Use /tools to see pending servers.

Model: Auto (/model to change)

> /agent list

* kiro_default
  default
  devops-agent
  orchestrator-agent
  cloud-operations-specialist
  jira-pm-assistant
  frontend-agent
  migration-modernization-specialist
  cloud-security-compliance-officer
  security-guardian
  dev-workshop
  solution-architect
  enterprise-cloud-architect
  dba-agent
  cloud-financial-analyst
  platform-engineering-lead
  fullstack-developer
  conductor
  qa-agent

>
```

cloudar

Power, flexibility, and security

Familiar environments

Kiro imports VS Code settings in a streamlined AI-ready environment and is also now available in the terminal.

Cutting-edge models

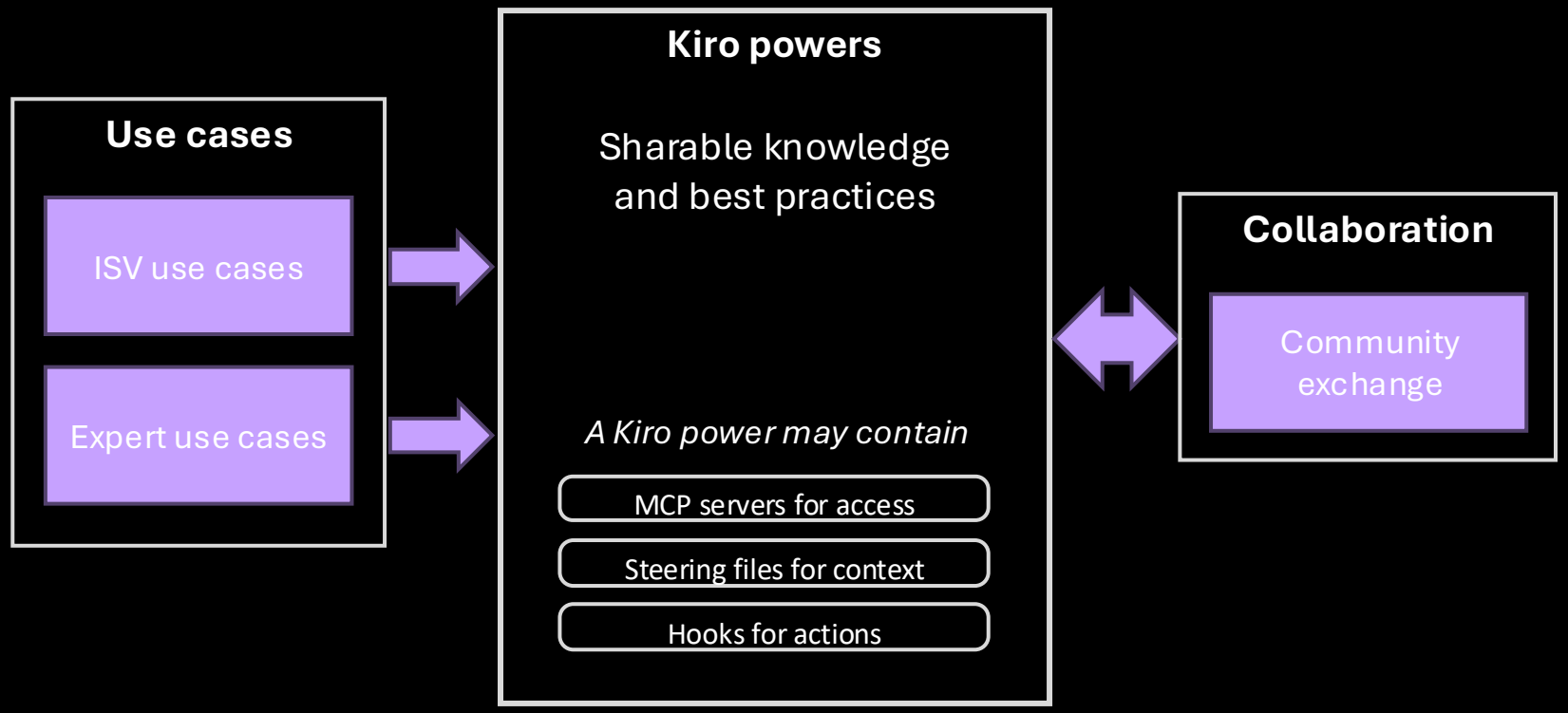
Choose from Claude Sonnet 4, 4.5 or Haiku 4.5 models, with Auto available for intelligent routing.

Enterprise-grade security

Kiro is built and operated by AWS

Introducing Kiro powers

Extend Kiro agent capabilities through sharable best practices



Kiro powers – Benefits

Targeted context for fast decisions

- Dynamic loading to avoid context overload
- One click download for IDE use
- Option to customize a Kiro power

Knowledge from ISVs and experts

- Best practices from ISVs for dev tools
- Opinionated guidance from experts
- Mechanism to share powers with the community

Dev to deployment use cases

- Application development spanning fullstack use cases
- Automated code build and deployment
- Observability on application performance and security

What if Kiro didn’t just help you work — but learned how you work and then able to independently solve problems for you — with your oversight?

Introducing Kiro autonomous agent

Kiro Web Portal

app.kiro.dev/agent/sessions/6047f611-a202-4d9e-b77a-4fa4d532f284

KIRO Autonomous agent (Preview) | Account

email@gmail.com K

Home

Chats

Tasks

Settings

We shouldn't have dependencies on the timely and d...

Cancel task

3:30 PM

I'm working on moving the timely and differential-dataflow dependencies from the hydro repository to the hydro-deps repository while preserving benchmark functionality.

3:48 PM

I'm performing final verifications and then I'll raise this for review.

3:53 PM

I've created the following pull requests for you to review:

- https://github.com/kyleandmikescrazyadventure/hydro/pull/100
- https://github.com/kyleandmikescrazyadventure/hydro-deps/pull/17

Update the task or ask questions...

We shouldn't have dependencies on the timely and d...

Copy ID

Completed

Repos: kyleandmikescrazyadventure/hydro, kyleandmikescrazyadventure/hydro-deps

PR #100 PR #17

Show detailed view

Creating task – Nov 30, 3:18 PM

Working on task

Creating pull request(s) PR #100 PR #17

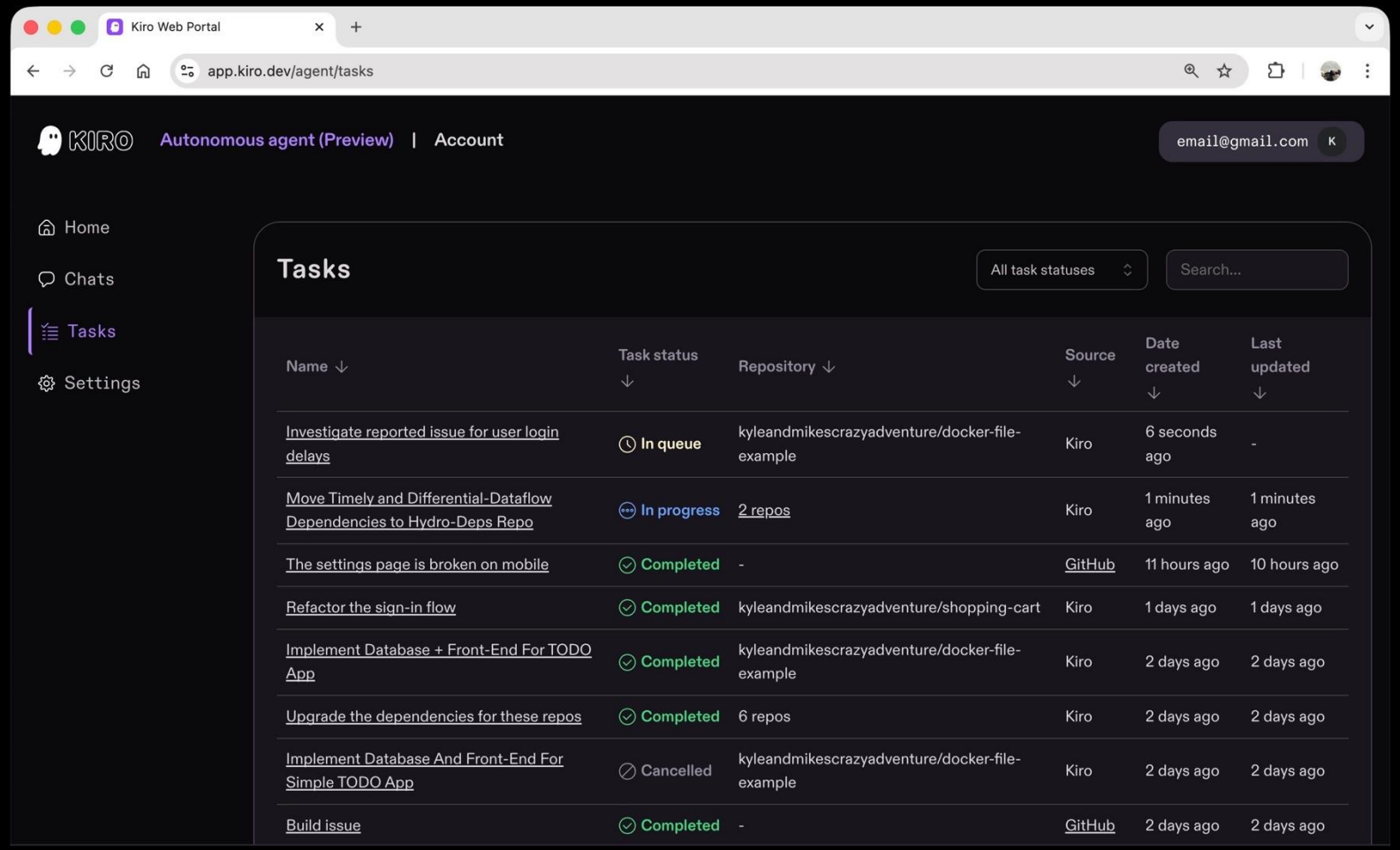
Completed

The autonomous agent that extends your flow in Kiro, now available in gated preview for both individuals and teams.

cloudar

Parallel execution without blocking your flow

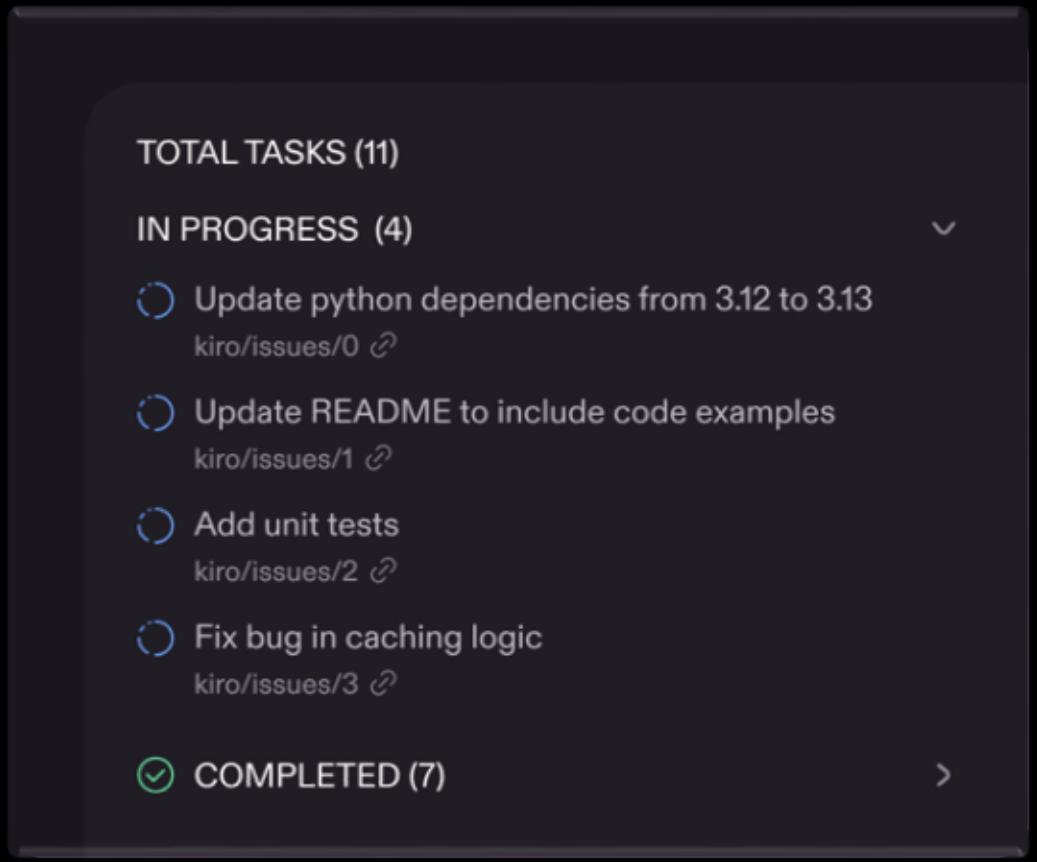
Chat with it, describe a change you need, or an improvement you want, and execute up to 10 tasks concurrently. It independently figure out how to get the work done.



Context that stays with your work

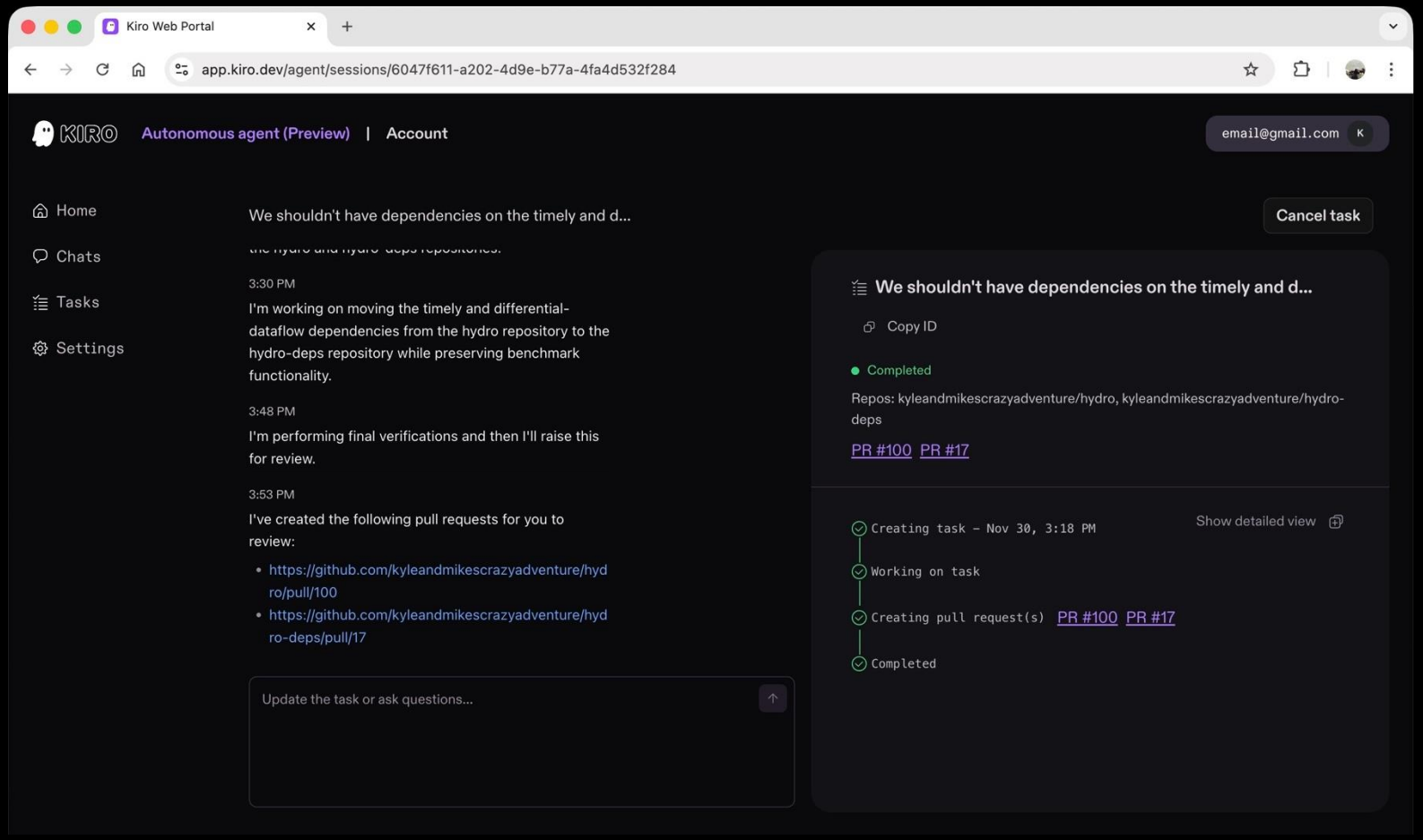
Kiro autonomous agent is always there and maintains context across your work.

- Kiro remembers feedback and applies that pattern to subsequent change.
- When Kiro encounters similar architectural decisions, it considers existing implementations and preferences.
- You're not re-explaining your codebase or repeating the same work—Kiro already knows how you work and gets better with each interaction.



Multi-repo changes in one move

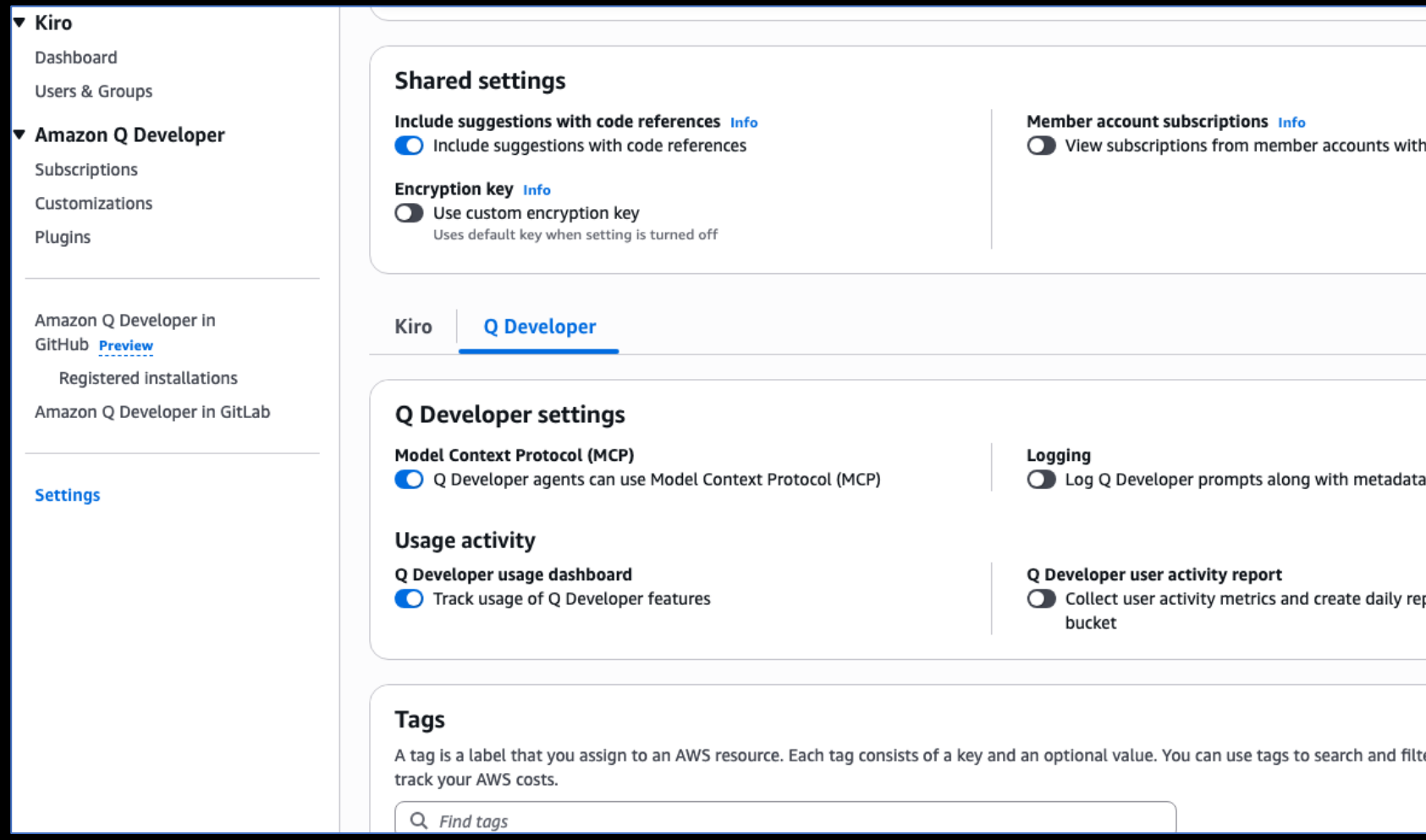
Describe the job once. Kiro treats the multi-repo work as a unified task--identifying the affected repo, analyzing how each service uses the library, updating code following your patterns, running full test suites, and opens pull-requests for review, all while you work on something else.



KIRO

Set up Kiro for your teams

- Understand and track usage with the advanced dashboards
- Manage settings with administrative controls
- Subscriptions available in AWS IAM Identity Center



cloudar

Pricing

Access the Kiro IDE and CLI with one subscription:

$20 /month	$40 /month	$200 /month
Pro Tier	**Pro+ Tier**	**Power Tier**
1,000 credits	2,000 credits	10,000 credits

Pay-per-use overages available for paid plans at $0.04 per additional credit

*Free Tier available with 50 credits per month, available for social-sign ins.

Get started with Kiro for free

Kiro.dev

cloudar

Thank you